PRANJAL SINGH

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SUMMARY

Final-year BCA (AI & Data Science) student with hands-on experience in machine learning pipelines, document classification systems, and backend REST APIs. Built TradeGuard (document intelligence & risk-scoring engine), completed two software engineering internships, and contributed 5 merged PRs to open-source project Kestra. Seeking a Machine Learning Intern role to apply practical skills in OCR, NLP, and production AI systems.

EDUCATION

Graphic Era Hill University — B.C.A., Artificial Intelligence & Data Science (CGPA: 7.82) **Jun 2024 – Jun 2027**

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), C, C++, SQL
ML & Data:	scikit-learn, XGBoost, k-NN, TF-IDF, NLP (basics), OCR (OpenCV, pypdf), NumPy, pandas
Web & APIs:	FastAPI, Flask, Node.js, Express.js, React.js, WebSockets, REST APIs, JWT Auth
Databases & Tools:	PostgreSQL, MongoDB, SQLite, SQLAlchemy, Git, GitHub, Postman, Linux, VS Code

EXPERIENCE

Full Stack Developer Intern — Technology Business Incubator (TBI), GEHU **Jun 2026 – Aug 2026**

AgriConnect | [GitHub] [Live Demo] · React.js, Next.js, Node.js, Express.js, MongoDB, Google Gemini API

- Built REST APIs for farmer-buyer agricultural marketplace with JWT authentication, bcrypt password hashing, and rate limiting.
- Integrated Google Gemini API using prompt engineering to enforce structured JSON responses for automated farm advisory.

Backend Developer Intern — Uptoskills (Remote) **May 2026 – Aug 2026**

LMS Platform · Node.js, Express.js, MongoDB, JWT

- Developed Interview Management Module with 6 REST APIs for scheduling and records, **reducing candidate processing time by 30%**.
- Audited 20+ backend routes, resolving 8 broken endpoints to improve platform reliability and error-handling.

PROJECTS

TradeGuard — Document Verification & Risk-Scoring Engine **Sep 2026 – Oct 2026**

[GitHub] · Python, FastAPI, XGBoost, scikit-learn, OpenCV, SQLAlchemy, SQLite

- Built TF-IDF classifier to categorize cargo documents (Bill of Lading, Invoice, Packing List) with regex extraction, ISO 6346 container check, and cross-document reconciliation for cargo weight and package validation.
- Trained XGBoost risk model on 500 synthetic shipments achieving **ROC-AUC 0.86** and **100% precision** on auto-release; exposed via FastAPI with JWT auth, interactive HTML dashboard, and 23 passing tests.

NearShare — E2EE P2P File Transfer with ML-Driven Optimization **Jul 2025 – Sep 2025**

[GitHub] · React 19, Node.js, WebSockets, SQLite, AES-256-GCM, ECDH, k-NN ML Model

- Engineered local & remote P2P file sharing with zero-cloud storage, securing chunk streams via **ECDH (`prime256v1`)** ephemeral key negotiation, **AES-256-GCM** authenticated encryption, and SHA-256 integrity digests.
- Implemented an embedded **k-NN telemetry model** learning from session metrics (payload size, peer count) to dynamically optimize chunk sizing (256KB–2MB) and bypass redundant compression on media formats.

StructDB — Custom Relational Database Engine **Jan 2025 – Apr 2025**

[GitHub] · Python, Custom SQL Parser, B-Tree, SHA-256, Tkinter

- Implemented custom DBMS from scratch featuring SQL parser, B-Tree indexing for O(log n) lookups, ACID transactions with rollback, SHA-256 hashed storage, and a Tkinter GUI for visual query execution.

OPEN SOURCE & ACHIEVEMENTS

Kestra — Open-source orchestration platform | [5 Merged PRs] · [15k+ ⭐ GitHub] 2026

- Merged **5 pull requests**: refactored frontend UI components, wrote automated unit tests, and resolved issue-tracked bugs.
- Solved **200+ problems on LeetCode** (leetcode.com/u/Pranjasinghl13) covering DSA, Dynamic Programming, and Graphs.
- Placed in **Top 10** at the **College Internal Hackathon**.